CSA 0921 - PROGRAMMING IN JAVA
CO2 - AT1 - SCENARIO BASED

NAME : VIDHUSHINI.T.S

REG. NO.: 192511061

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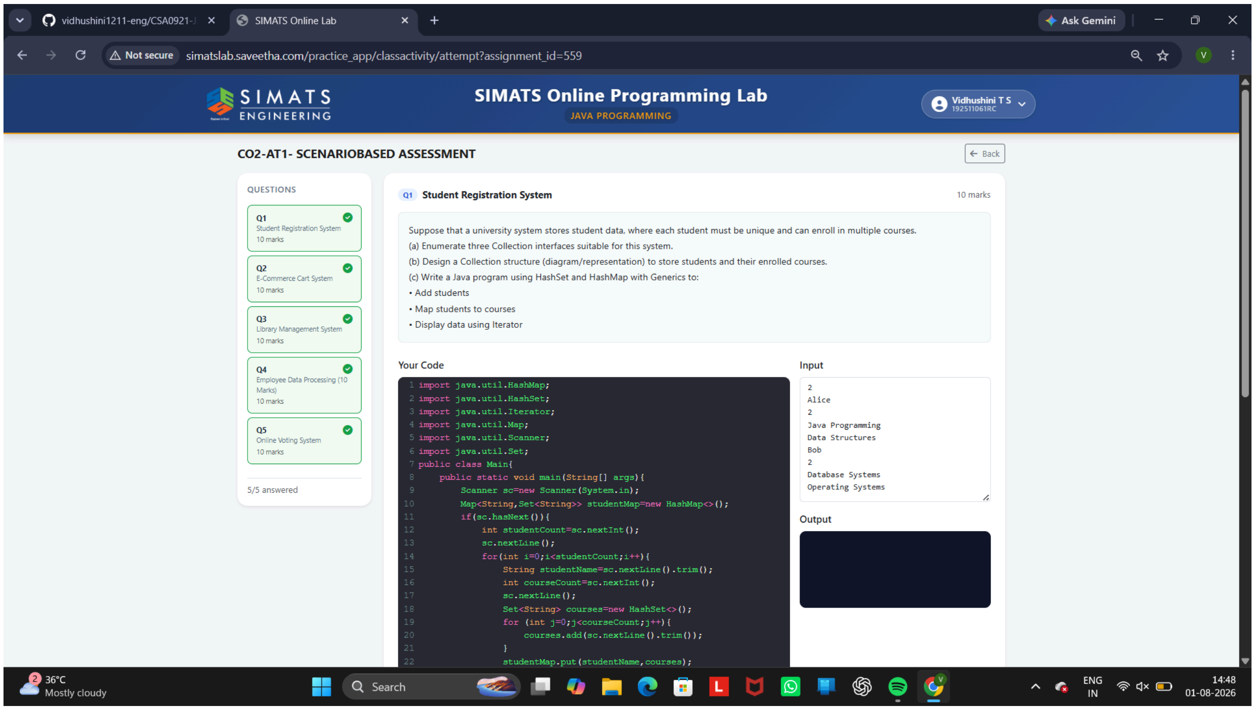

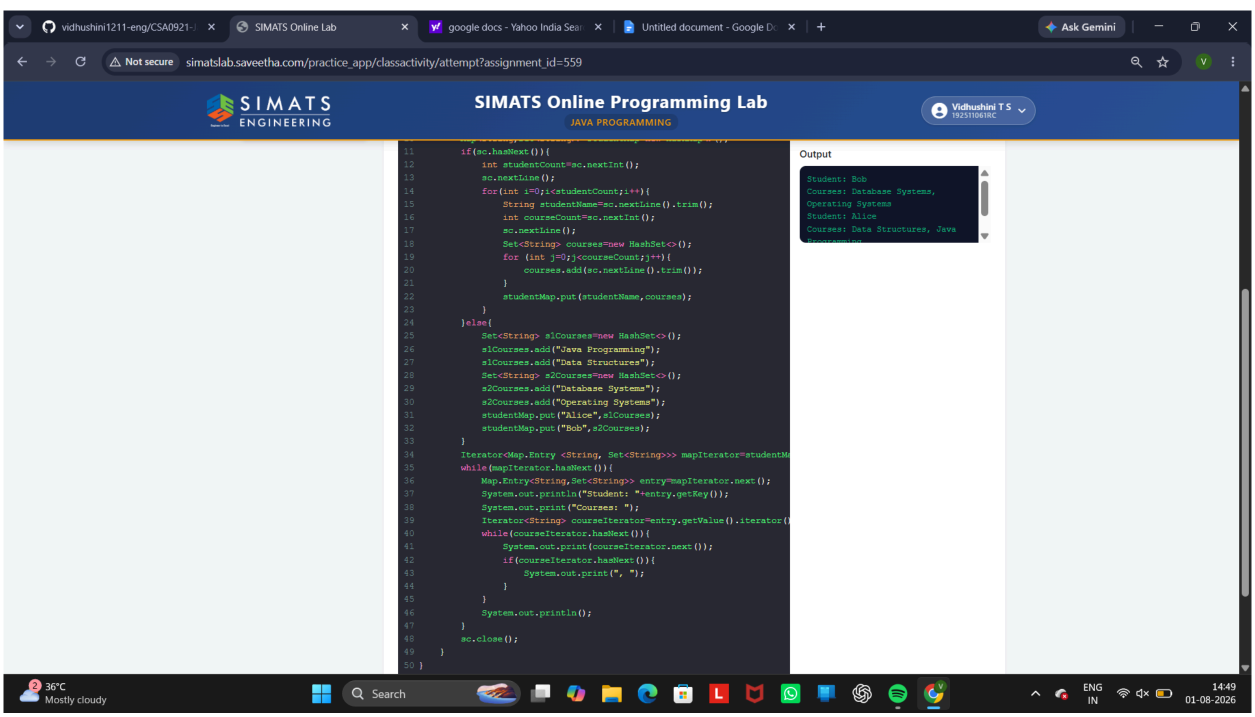

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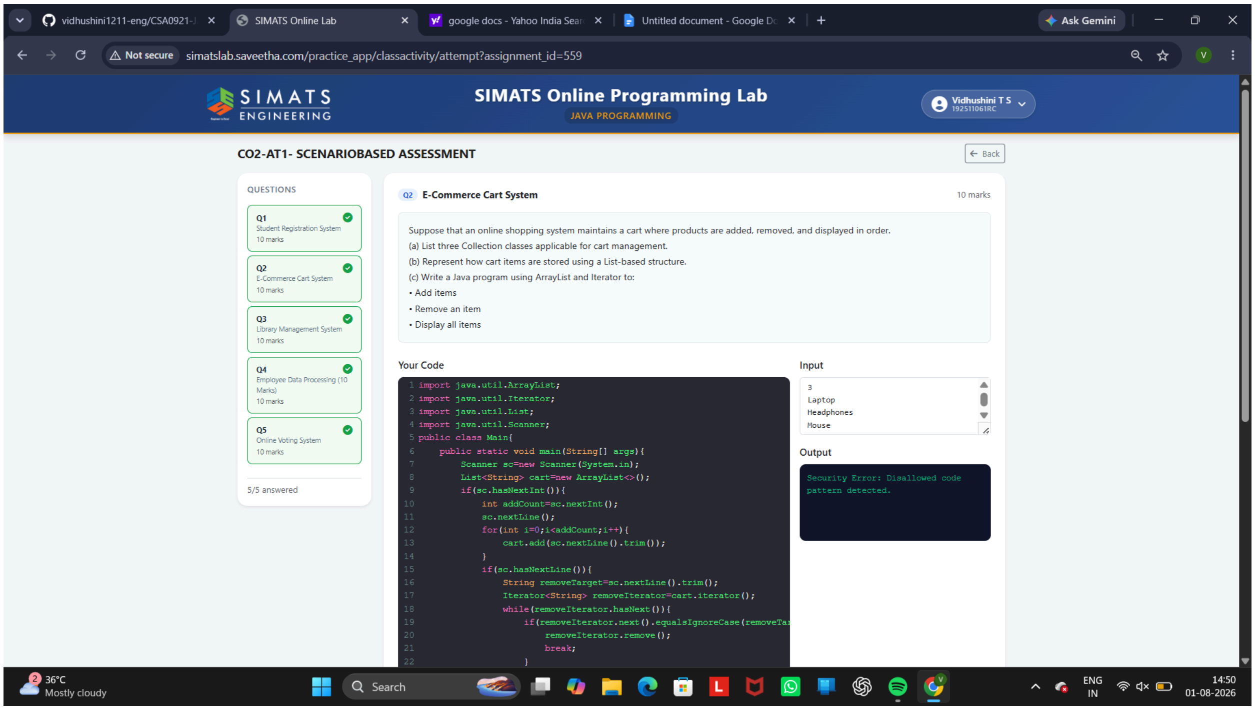

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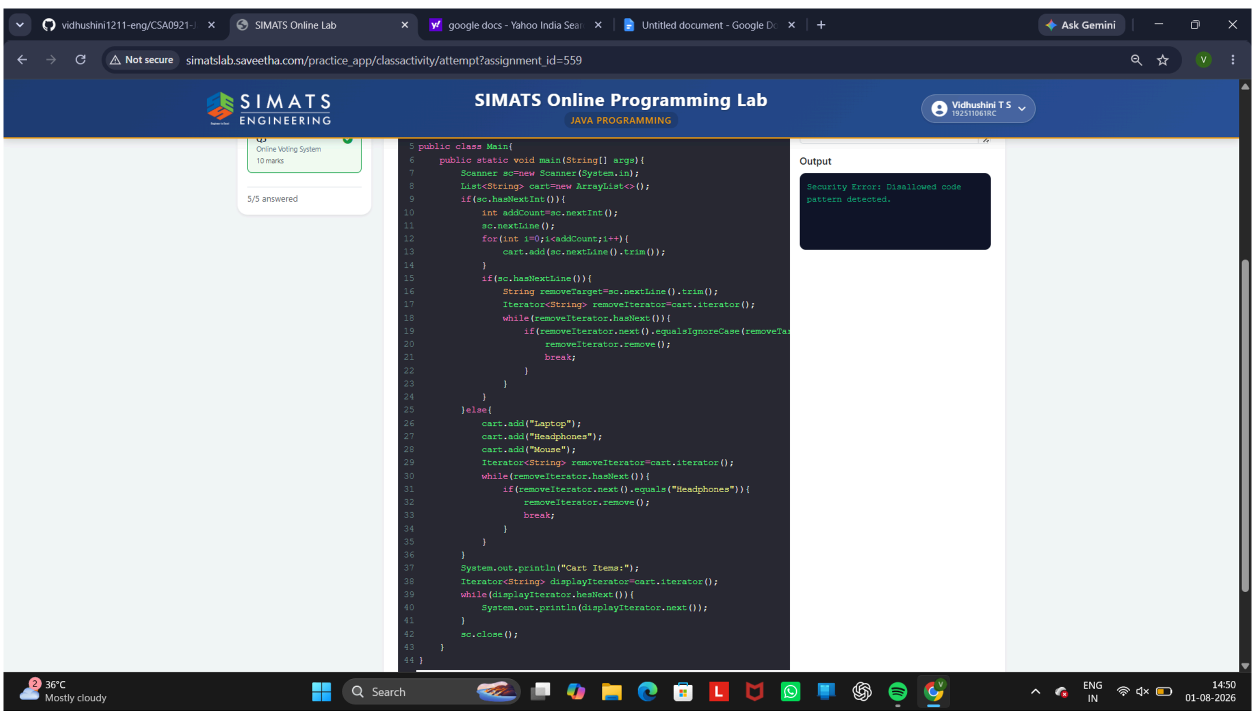
CO2-AT1- SCENARIOBASED ASSESSMENT
Back
QUESTIONS
Q1
Student Registration System
10 marks
Q2
E-Commerce Cart System
10 marks
Q3
Library Management System
10 marks
Q4
Employee Data Processing (10 Marks)
10 marks
Q5
Online Voting System
10 marks
5/5 answered
Q2
E-Commerce Cart System
10 marks
Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.
(a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:
• Add items
• Remove an item
• Display all items
Your Code
import java.util.ArrayList;
import java.util.Iterator;
import java.util.List;
import java.util.Scanner;
public class Main{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
List<String> cart=new ArrayList<>();
if(sc.hasNextInt()){
int addCount=sc.nextInt();
sc.nextLine();
for(int i=0;i<addCount;i++){
cart.add(sc.nextLine().trim());
}
if(sc.hasNextLine()){
String removeTarget=sc.nextLine().trim();
Iterator<String> removeIterator=cart.iterator();
while(removeIterator.hasNext()){
if(removeIterator.next().equalsIgnoreCase(removeTa
removeIterator.remove();
break;
}
Input
3
Laptop
Headphones
Mouse
Output
Security Error: Disallowed code pattern detected.
Search
36°C
Mostly cloudy
ENG
IN
14:50
01-08-2026



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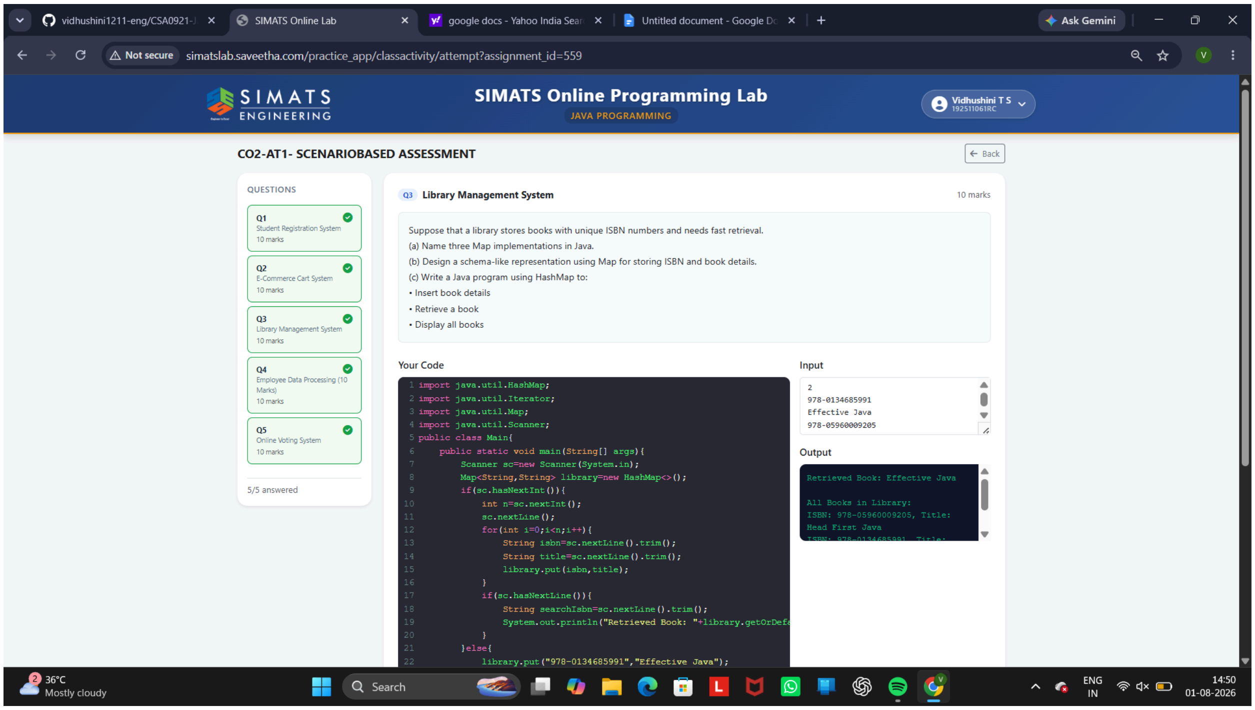
Online Voting System
10 marks
5/5 answered
public class Main{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
List<String> cart=new ArrayList<>();
if(sc.hasNextInt()){
int addCount=sc.nextInt();
sc.nextLine();
for(int i=0;i<addCount;i++){
cart.add(sc.nextLine().trim());
}
if(sc.hasNextLine()){
String removeTarget=sc.nextLine().trim();
Iterator<String> removeIterator=cart.iterator();
while(removeIterator.hasNext()){
if(removeIterator.next().equalsIgnoreCase(removeTa
removeIterator.remove();
break;
}
}
}
}else{
cart.add("Laptop");
cart.add("Headphones");
cart.add("Mouse");
Iterator<String> removeIterator=cart.iterator();
while(removeIterator.hasNext()){
if(removeIterator.next().equals("Headphones")){
removeIterator.remove();
break;
}
}
}
System.out.println("Cart Items:");
Iterator<String> displayIterator=cart.iterator();
while(displayIterator.hesNext()){
System.out.println(displayIterator.next());
}
sc.close();
}
}
Output
Security Error: Disallowed code pattern detected.
Search
36°C
Mostly cloudy
ENG
IN
14:50
01-08-2026

3.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Back
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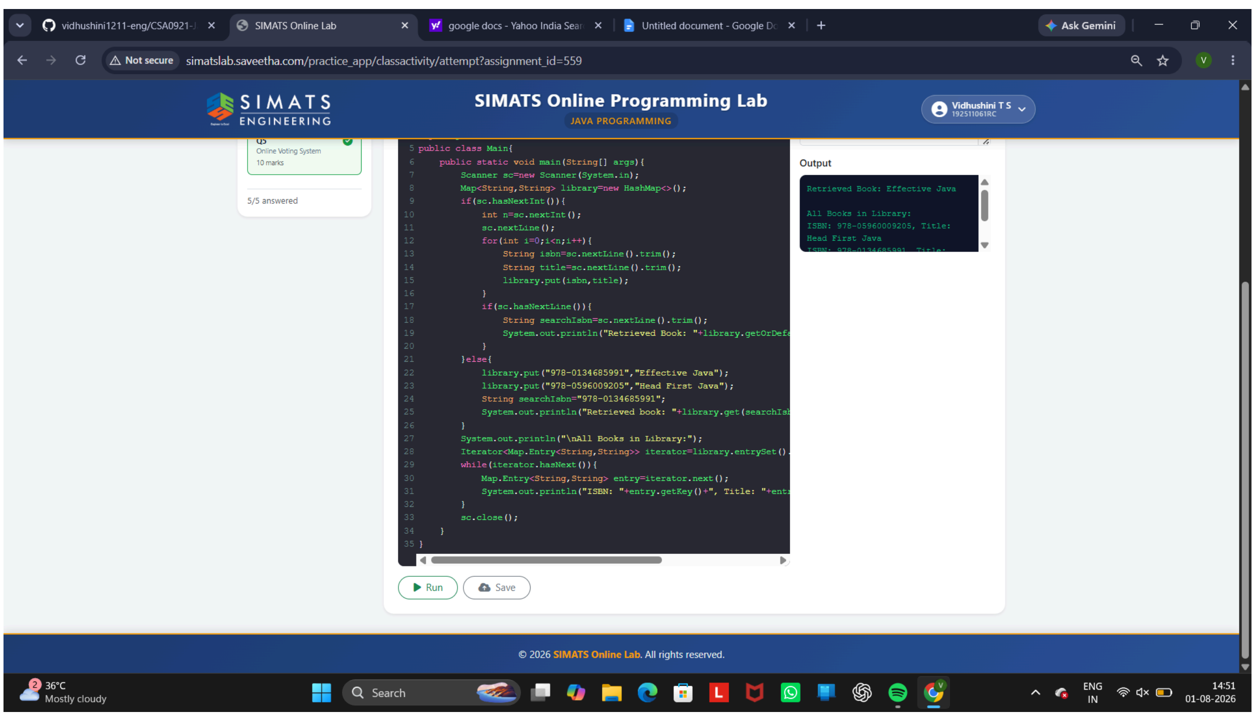
Q2 E-Commerce Cart System 10 marks
Q3 Library Management System 10 marks
Q4 Employee Data Processing (10 Marks) 10 marks
Q5 Online Voting System 10 marks
5/5 answered
Q3 Library Management System
10 marks
Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.
(a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:
• Insert book details
• Retrieve a book
• Display all books
Your Code
import java.util.HashMap;
import java.util.Iterator;
import java.util.Map;
import java.util.Scanner;
public class Main{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
Map<String,String> library=new HashMap<>();
if(sc.hasNextInt()){
int n=sc.nextInt();
sc.nextLine();
for(int i=0;i<n;i++){
String isbn=sc.nextLine().trim();
String title=sc.nextLine().trim();
library.put(isbn,title);
}
if(sc.hasNextLine()){
String searchIsbn=sc.nextLine().trim();
System.out.println("Retrieved Book: "+library.getOrDefa
}
}else{
library.put("978-0134685991","Effective Java");
Input
2
978-0134685991
Effective Java
978-05960009205
Output
Retrieved Book: Effective Java
All Books in Library:
ISBN: 978-05960009205, Title:
Head First Java
36°C
Mostly cloudy
Search
ENG
IN
14:50
01-08-2026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Online Voting System
10 marks
5/5 answered
public class Main{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
Map<String,String> library=new HashMap<>();
if(sc.hasNextInt()){
int n=sc.nextInt();
sc.nextLine();
for(int i=0;i<n;i++){
String isbn=sc.nextLine().trim();
String title=sc.nextLine().trim();
library.put(isbn,title);
}
if(sc.hasNextLine()){
String searchIsbn=sc.nextLine().trim();
System.out.println("Retrieved Book: "+library.getOrDefa
}
}else{
library.put("978-0134685991","Effective Java");
library.put("978-0596009205","Head First Java");
String searchIsbn="978-0134685991";
System.out.println("Retrieved book: "+library.get(searchIsb
}
System.out.println("\nAll Books in Library:");
Iterator<Map.Entry<String,String>> iterator=library.entrySet().
while(iterator.hasNext()){
Map.Entry<String,String> entry=iterator.next();
System.out.println("ISBN: "+entry.getKey()+", Title: "+entr
}
sc.close();
}
}
Run
Save
Output
Retrieved Book: Effective Java
All Books in Library:
ISBN: 978-05960009205, Title:
Head First Java
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Search
ENG
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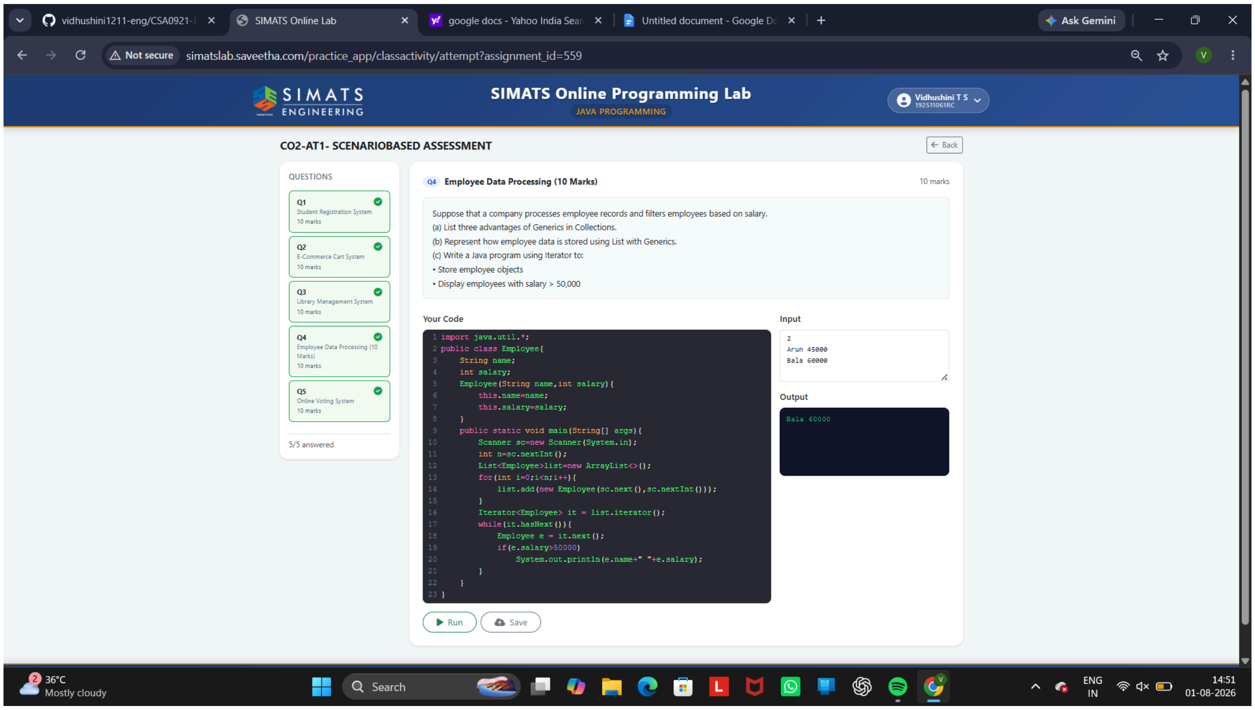
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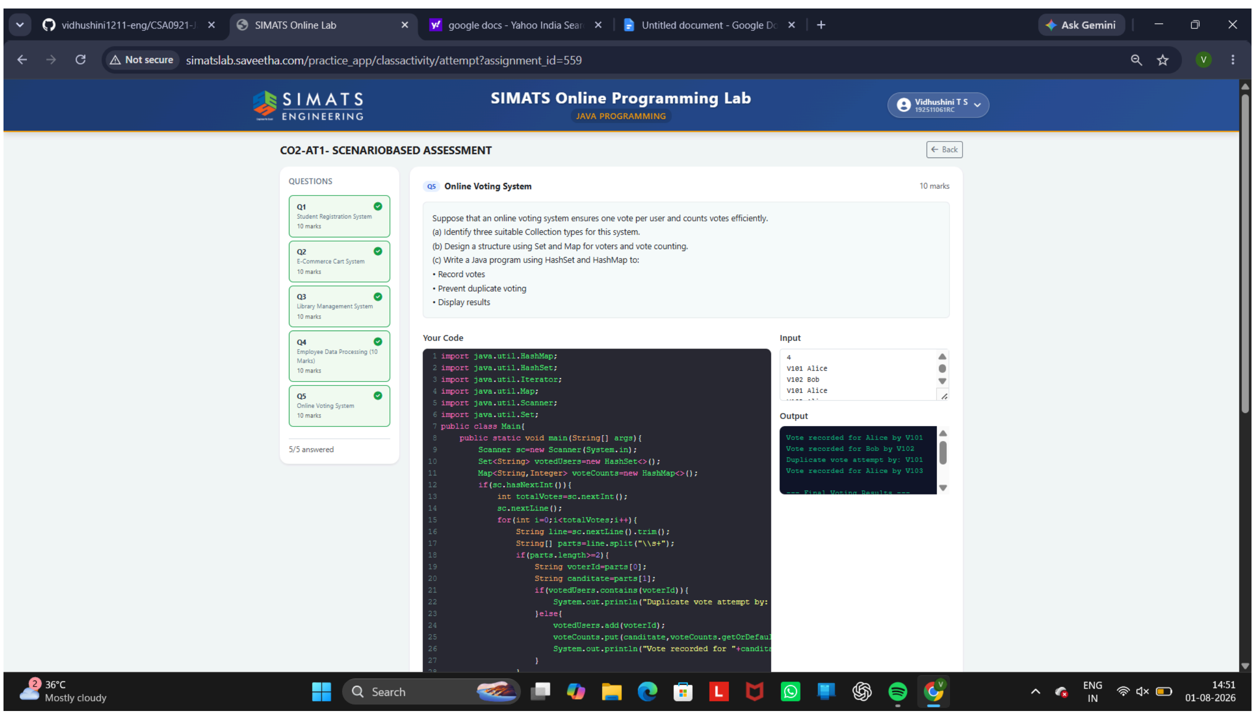
Q5 Online Voting System 10 marks
5/5 answered
Q4 Employee Data Processing (10 Marks)
10 marks
Suppose that a company processes employee records and filters employees based on salary.
(a) List three advantages of Generics in Collections.
(b) Represent how employee data is stored using List with Generics.
(c) Write a Java program using Iterator to:
• Store employee objects
• Display employees with salary > 50,000
Your Code
import java.util.*;
public class Employee{
String name;
int salary;
Employee(String name,int salary){
this.name=name;
this.salary=salary;
}
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
List<Employee>list=new ArrayList<>();
for(int i=0;i<n;i++){
list.add(new Employee(sc.next(),sc.nextInt()));
}
Iterator<Employee> it = list.iterator();
while(it.hasNext()){
Employee e = it.next();
if(e.salary>50000)
System.out.println(e.name+" "+e.salary);
}
}
}
Input
2
Arun 45000
Bala 60000
Output
Bala 60000
Run
Save

5.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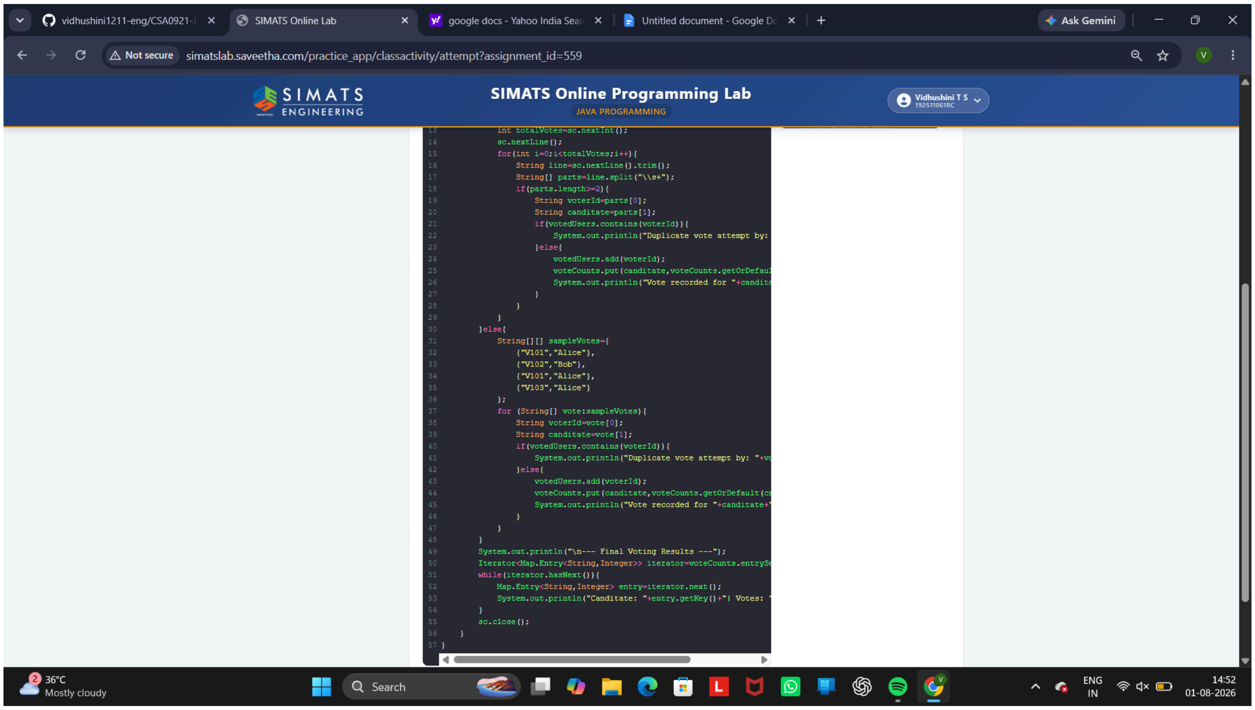
Q5 Online Voting System 10 marks
5/5 answered
Q5 Online Voting System
10 marks
Suppose that an online voting system ensures one vote per user and counts votes efficiently.
(a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
• Record votes
• Prevent duplicate voting
• Display results
Your Code
import java.util.HashMap;
import java.util.HashSet;
import java.util.Iterator;
import java.util.Map;
import java.util.Scanner;
import java.util.Set;
public class Main{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
Set<String> votedUsers=new HashSet<>();
Map<String,Integer> voteCounts=new HashMap<>();
if(sc.hasNextInt()){
int totalVotes=sc.nextInt();
sc.nextLine();
for(int i=0;i<totalVotes;i++){
String line=sc.nextLine().trim();
String[] parts=line.split("\\s+");
if(parts.length>=2){
String voterId=parts[0];
String canditate=parts[1];
if(votedUsers.contains(voterId)){
System.out.println("Duplicate vote attempt by:
}else{
votedUsers.add(voterId);
voteCounts.put(canditate,voteCounts.getOrDefaul
System.out.println("Vote recorded for "+candita
}
Input
4
V101 Alice
V102 Bob
V101 Alice
Output
Vote recorded for Alice by V101
Vote recorded for Bob by V102
Duplicate vote attempt by: V101
Vote recorded for Alice by V103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Vidhushini T S
int totalVotes=sc.nextInt();
sc.nextLine();
for(int i=0;i<totalVotes;i++){
String line=sc.nextLine().trim();
String[] parts=line.split("\\s+");
if(parts.length>=2){
String voterId=parts[0];
String canditate=parts[1];
if(votedUsers.contains(voterId)){
System.out.println("Duplicate vote attempt by:
}else{
votedUsers.add(voterId);
voteCounts.put(canditate,voteCounts.getOrDefaul
System.out.println("Vote recorded for "+candita
}
}
}
}else{
String[][] sampleVotes={
{"V101","Alice"},
{"V102","Bob"},
{"V101","Alice"},
{"V103","Alice"}
};
for (String[] vote:sampleVotes){
String voterId=vote[0];
String canditate=vote[1];
if(votedUsers.contains(voterId)){
System.out.println("Duplicate vote attempt by: "+vo
}else{
votedUsers.add(voterId);
voteCounts.put(canditate,voteCounts.getOrDefault(ca
System.out.println("Vote recorded for "+canditate+"
}
}
}
System.out.println("\n--- Final Voting Results ---");
Iterator<Map.Entry<String,Integer>> iterator=voteCounts.entrySe
while(iterator.hasNext()){
Map.Entry<String,Integer> entry=iterator.next();
System.out.println("Canditate: "+entry.getKey()+"| Votes: "
}
sc.close();
}
}
36°C
Mostly cloudy
Search
ENG
IN
14:52
01-08-2026